

SHINE DANIEL

Kochi, Kerala, India

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Career Objective

To pursue a career in Electronics and Communication Engineering where I can apply my hands-on experience in hardware development, embedded systems, and project-based engineering. I aim to contribute to innovative technical solutions while continuously developing my expertise in industrial automation and digital electronics.

Education

Bachelor of Technology in Electronics and Communication Engineering

Rajagiri School of Engineering and Technology, Kochi

2023 -- 2027

CGPA: 9.21 / 10

Class XII, CBSE

Indian School, Ras Al Khaimah (RAK), UAE

Graduated 2023 | 95%

Class X, CBSE

Indian School, Ras Al Khaimah (RAK), UAE

Graduated 2021 | 94%

Technical Skills

- **Programming Languages:** Python, Embedded C, Java, HTML, CSS, Assembly Language
- **HDL & Digital Design:** Verilog HDL, SystemVerilog, Cadence Virtuoso
- **Embedded Systems and Hardware:** ESP32, Arduino Uno, Raspberry Pi Pico, Raspberry Pi, PLCs (Ladder Logic Programming), Analog ICs, Sensors, UART
- **Tools and Frameworks:** MATLAB, Arduino IDE, Vivado, Gazebo, RViz, ROS, MySQL, Linux, Autodesk Fusion 360, GNU Radio, KiCad, ANSYS HFSS, Keil uVision5, Studio 5000
- **Domains:** Embedded Systems, Robotics, Analog and Digital Circuit Design, IoT, Web Development, Industrial Automation, PCB Design, VLSI

Projects

Main Project: Seizure Detection in an Edge Device Using EEG

Raspberry Pi 5, ADS1299

- Developing a lightweight TCN model using depthwise-separable convolutions with a reduced parameter count for seizure detection.

8-bit ALU Design and Simulation

Cadence Virtuoso, VLSI

- Designed an 8-bit ALU with flag functionality and began implementing the design from RTL toward GDS using Cadence Virtuoso.

LUMINO - Spotify Controller Lamp

IoT, Embedded Electronics

- Designed and prototyped a desktop pencil-holder lamp with controller buttons/encoders for Spotify playback control and an integrated Pomodoro timer.
- Work in progress : electronics is done, now 3D modeling the body.

WePOD - ESP32 Walkman

Bluetooth & Audio Electronics

- Developing an ESP32-based Walkman that supports three playback modes: wired headphones, Bluetooth, and speaker output, along with a SPI display to show real-time playback.
- Work in progress — currently on hold due to component availability.

2-Band Audio Spectrum Analyser

Analog Electronics, Active Filters

- Designed a two-band audio spectrum analyser using active RC filter networks to visualise frequency content in real-time (100 Hz and 2.7 kHz).

4-Bit Full Adder Using Only Transistors

BC547, Discrete Components

- Built a 4-bit full adder entirely from discrete BC547 transistors, implementing the required logic gates at the transistor level.

Bluetooth-Controlled RC Car with LCD Display

Arduino, ESP32, Bluetooth

- Developed a Bluetooth-controlled RC car with live telemetry displayed on an onboard LCD screen. Implemented motor control, serial communication, and real-time display updates within the Arduino firmware.

Transistor-Level D Flip-Flop Design

Cadence Virtuoso, VLSI

- Designed and simulated a D flip-flop at the transistor level using Cadence Virtuoso for digital sequential circuit analysis.

Robotic Arm Simulation in Gazebo and RViz

ROS, Gazebo, RViz

- Modeled and simulated a 6-DOF multi-joint robotic arm using ROS, Gazebo, and RViz for motion planning and 3D visualisation.

Frequency Multiplier using CD4046 PLL

CD4046 IC, Analog Design

- Designed a frequency multiplier circuit using the phase-locked loop (PLL) functionality of the CD4046 IC.

Ouroboros: Bio-Inspired Snake Robot

Raspberry Pi Pico, ESP32, MPU6050

- Developed a bio-inspired helical snake robot for autonomous navigation in confined and irregular terrains, featuring three articulated segments and five geared motors with integrated IMU sensing. Implemented UART-based control and a web dashboard for real-time monitoring and maneuverability.

Experience

Summer Internship

May 2025 -- June 2025

Magnum Technology Center, JAFZA, Dubai

- Gained hands-on experience with PLC programming using Studio 5000, Logix Emulate and industrial automation workflows in a professional manufacturing environment.
- Developed practical understanding of industrial safety standards and real-world control system design and operation.

GIS Mapathon

Jan 2026 -- Feb 2026

RSET in collaboration with NeST Digital

- Deployed and trained a U-Net model on a local GPU for pavement-marking extraction, then visualised the output as an overlay in QGIS. Placed in the top 9 out of 100 teams.
- Placed first in the Story Mapping sub-competition.
- Developed practical experience in AI model deployment and real-world problem-solving workflows.

CarbonX - HackS'US Edition V

Mar 2026

ECE Dept in collaboration with C-DAC

- Served as Track Head for Vegathon, a 48-hour Vegaboard-focused hackathon.
- Served as the point of contact between participating teams and C-DAC.

Leadership & Extracurricular Activities

Technical Coordinator / Student Co-Lead

Electronauts, ECE Club, RSET

- Organised and coordinated multiple technical events ranging from activity-hour sessions to tech-fest-level competitions.
- Coordinated events including Chips2Silicon, Blindbuild, Blindbuild 2.0, CodeQuest, CodeQuest 2.0, Wire It Right, PAYLOAD, and CarbonX (Vegathon).

Media Team Member

RSET Media Team

- Contributed to recording and documenting major college events including Bharatham, Confluence, and Onam celebrations.
- Developed team coordination and event documentation skills through consistent collaborative work.

Additional Courses

- Embedded Sensing, Actuation and Interfacing Systems | NPTEL
- Machine Learning for Engineering and Science Application | NPTEL
- Microprocessors and Microcontrollers | NPTEL

Core Skills

- Problem Solving | Team Collaboration | Event Coordination
- Communication Skills | Initiative and Self-Learning | Time Management

Languages

- English | Malayalam | Hindi